

# ORGNZ-05: Bucket-Level Access Control

**Series:** ORGNZ | **Notebook:** 5 of 9 | **Created:** January 2026 | **Last Updated:** 01/28/2026

## Overview

Bucket-level access control provides a straightforward way to isolate data by team, application, or business unit. By granting permissions to specific buckets, you can ensure teams only access data relevant to their responsibilities.

## Prerequisites

| Requirement | Details |  
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| **Dynatrace Account** | Account-level administrative access |

| **Permissions** | IAM policy management permissions |

| **Knowledge** | Completed ORGNZ-04 |

## Learning Objectives

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By the end of this notebook, you will:

- Create IAM policies for bucket-level access
- Use bucket naming patterns in policies
- Implement team isolation using buckets
- Understand bucket permission best practices

## Bucket Permission Fundamentals

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### Required Permissions

All bucket access policies must start with `storage:buckets:read` :

```
ALLOW storage:buckets:read WHERE <condition>;  
ALLOW storage:logs:read; // Then allow table access
```

### Two-Step Pattern

| Step             | Purpose                 | Example                                                   |
|------------------|-------------------------|-----------------------------------------------------------|
| 1. Bucket access | Define which buckets    | <code>storage:buckets:read WHERE bucket-name = 'x'</code> |
| 2. Table access  | Define which data types | <code>storage:logs:read , storage:metrics:read</code>     |

# Policy Examples

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## Example 1: Single Bucket Access

Grant a team access to their specific bucket:

```
{
  "name": "platform-team-logs-access",
  "description": "Platform team can access platform logs bucket",
  "statementQuery": "ALLOW storage:buckets:read WHERE storage:bucket-name = 'team_platform_logs'; ALLOW storage:logs:read;",
  "tags": ["team:platform"]
}
```

## Example 2: Multiple Buckets with IN Operator

Grant access to a list of specific buckets:

```
{
  "name": "finance-multi-bucket-access",
  "description": "Finance team can access multiple buckets",
  "statementQuery": "ALLOW storage:buckets:read WHERE storage:bucket-name IN ('finance_logs', 'finance_metrics', 'finance_audit'); ALLOW storage:logs:read, storage:metrics:read;",
  "tags": ["team:finance"]
}
```

## Example 3: Bucket Name Pattern with STARTSWITH

Grant access to all buckets matching a prefix:

```
{
  "name": "prod-infrastructure-access",
  "description": "Access to all production infrastructure buckets",
  "statementQuery": "ALLOW storage:buckets:read WHERE storage:bucket-name STARTSWITH 'prod_infra_'; ALLOW storage:logs:read, storage:metrics:read, storage:spans:read;",
  "tags": ["env:production"]
}
```

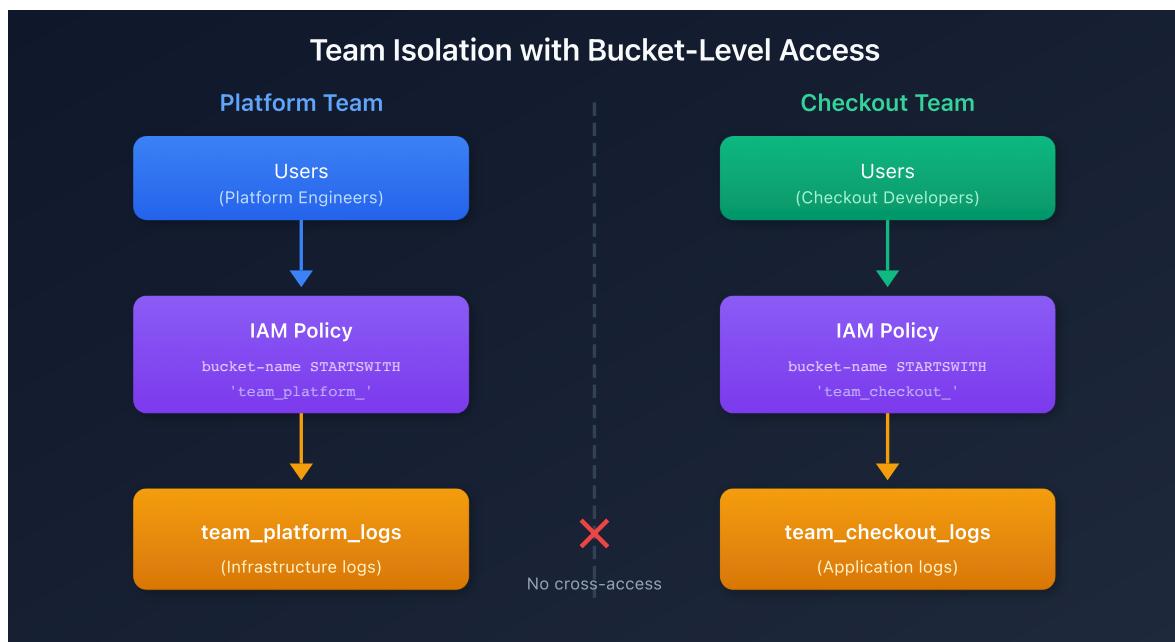
## Example 4: Bucket Pattern with MATCH

Use pattern matching for complex bucket names:

```
{
  "name": "database-team-access",
  "description": "Database team can access any database-related bucket",
  "statementQuery": "ALLOW storage:buckets:read WHERE storage:bucket-name
MATCH ('*-database-*'); ALLOW storage:logs:read;",
  "tags": ["team:database"]
}
```

## Team Isolation Pattern

### Architecture



### Implementation

#### Step 1: Create team buckets

```
team_platform_logs
team_checkout_logs
team_payments_logs
```

## Step 2: Route data via OpenPipeline

```
processors:
  - type: route
    rules:
      - condition: "host.group starts-with 'platform-'"
        destination: "team_platform_logs"
      - condition: "service.name contains 'checkout'"
        destination: "team_checkout_logs"
```

## Step 3: Create IAM policies per team

```
// Platform team policy
ALLOW storage:buckets:read WHERE storage:bucket-name STARTSWITH
"team_platform_";
ALLOW storage:logs:read;

// Checkout team policy
ALLOW storage:buckets:read WHERE storage:bucket-name STARTSWITH
"team_checkout_";
ALLOW storage:logs:read;
```

# Default Buckets Policy

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## Standard Default Access

For users who need access to all default buckets:

```
ALLOW storage:buckets:read WHERE storage:bucket-name STARTSWITH "default_";
ALLOW storage:events:read, storage:logs:read, storage:metrics:read,
storage:entities:read, storage:bizevents:read, storage:spans:read;
```

## Default Plus Custom Buckets

```
ALLOW storage:buckets:read WHERE storage:bucket-name STARTSWITH "default_";
ALLOW storage:buckets:read WHERE storage:bucket-name = "audit_logs_365d";
ALLOW storage:logs:read, storage:metrics:read, storage:spans:read;
```

# Verifying Bucket Access

```
// List buckets you have access to
fetch dt.system.buckets
| fields name, display_name, dt.system.table, retention_days
| sort name asc
```

```
// Query data from specific bucket to verify access
fetch logs, bucket: "team_platform_logs"
| limit 10
```

```
// Check data distribution across accessible buckets
fetch logs
| summarize count = count(), by:{dt.system.bucket}
| sort count desc
```

# Best Practices

| Practice                             | Rationale                            |
|--------------------------------------|--------------------------------------|
| Use consistent bucket naming         | Enables pattern-based policies       |
| Assign policies to groups, not users | Easier management, consistent access |
| Document all policies                | Audit trail and governance           |
| Test policies with sample users      | Prevent access issues                |
| Use STARTSWITH for team prefixes     | Future-proof for new buckets         |
| Combine with record-level for scale  | Bucket limits (80) may not suffice   |

# When Bucket-Level Isn't Enough

Bucket-level access has limitations:

| Limitation              | Alternative                                |
|-------------------------|--------------------------------------------|
| 80 bucket limit         | Use security context for finer granularity |
| Shared data needs       | Use record-level permissions               |
| Dynamic team membership | Use security context mapped to tags        |
| Field-level masking     | Use field-level permissions                |

See **ORGNZ-06** and **ORGNZ-07** for advanced permission patterns.

## Next Steps

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Continue with the ORGNZ series:

- **ORGNZ-06**: Security Context

## References

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- [Permissions in Grail](#)
  - [IAM policy reference](#)
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